

AMOL GUPTA

Senior Software Engineer — Full Stack (Java/React)

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years building payment platforms at Visa Inc. across React/TypeScript frontends and Java backend services. Deep expertise in backend and modern frontend architecture (Nx monorepos, TanStack Query/Router, React Hook Form, Zod, Tailwind CSS) alongside production backend ownership — designed and built a distributed feature-toggle platform on Vert.x and Hazelcast, and implemented search and payment-acceptance APIs over Solr, Cassandra and Spring. Drove 50% business growth and improved rendering performance by 55%.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Java, SQL, C++

Frontend: React 19, Redux/Redux Toolkit, TanStack Query/Router, React Hook Form, Zod, Tailwind CSS, Shadcn UI, HTML5/CSS3

Backend: Spring Boot, Vert.x, JAX-RS/Jersey, JPA/Hibernate, REST APIs, OAuth2/JWT, Maven

Data & Infrastructure: Apache Solr, Apache Lucene, Apache Cassandra, Hazelcast, Kafka, MySQL

Architecture & Testing: Nx monorepo, Distributed caching, Microservices, System Design, Agile/Scrum, Write-through cache synchronization, Multi-tenant authorization, Feature-flag infrastructure, PCI-DSS, Vitest, Playwright, Jest, JUnit, Mockito, Storybook

DevOps & Observability: Docker, Kubernetes, Helm, ArgoCD, Jenkins, CI/CD, Git, GitHub Enterprise, Splunk, Grafana, SonarQube, Checkmarx SAST

AI Tooling: GitHub Copilot, Claude Code, Visa MCP Hub

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Visa Inc.

Jun 2024 – Present

• APEX — Authorize.Net Merchant Portal (Frontend)

React 19, TypeScript, Vite, Nx, Redux, TanStack Query

- Led frontend for the Virtual Terminal and API Keys modules in a large Nx monorepo; architected the Manage Transactions module (real-time tracking, search, filtering), contributing to 50% business growth.
- Owned ~30% of the codebase and improved rendering performance by 55%; built shared component libraries cutting duplication across three squads
- Delivered a QuickBooks accounting integration and multi-currency support with locale-aware rendering for financial reconciliation across global markets with CSV batch/bulk upload interfaces, settlement and reconciliation reporting screens.

• Delphinus — Distributed Feature-Toggle Platform

Java, Vert.x, Hazelcast, MySQL, Helm, ArgoCD

- Designed and developed a **feature-flag** platform end to end — an embedded Java client library and a **Vert.x** backend service — giving internal services runtime rollout control, deployed as isolated per-region stacks for data residency.
- Architected a **four-tier** read path (Hazelcast grid → backend service → per-JVM cache → static config) keeping flag evaluation off the network in the common case and returning a usable value during a full backend outage, with write-through cache synchronization bounding staleness to a single write interval.
- Built deploy-time bulk registration with safety guardrails — in-file de-duplication, cache-based existence checks and a configurable batch cap (default **50**) — preventing a misconfigured deploy from writing unbounded flag state.
- Implemented group-based conditional rollout evaluated client-side to avoid centralizing **evaluation load**.

• Transaction Search Service

Java, Spring, Solr, Lucene, Cassandra, Kafka, React, Redux, JavaScript

- **Developed a transaction search** and analytics feature for searching, filtering, and exporting **billions** of historical payment records across a **four-level** organization hierarchy.
- Enforced multi-tenant access control at query construction — org-hierarchy filters compiled into the Solr query rather than applied as a post-filter — making cross-tenant results structurally unrepresentable.
- Designed Solr/Lucene search supporting simultaneous full-text, range and faceted queries, with deterministic pagination via a unique secondary sort key and bounded request cost (**2,500** rows per page, **100,000** max offset).
- Split storage by access pattern — Solr for search, Cassandra for saved searches and enrichment lookups with a **30-day** TTL on unsaved queries — keeping enrichment behind a feature toggle so store-level incidents degrade gracefully instead of failing search.
- Built the React/Redux search experience — autocomplete, filtering, result grids with server-driven pagination, and export.

Software Engineer — Visa Inc.

Jun 2022 – Jun 2024

• Accept Payment / Virtual Terminal Service

Java, Spring, OAuth2, JWT

- Developed a virtual terminal payment feature enabling agents to initiate real-time merchant payments through a multi-step workflow, supporting **six** transaction types across manual card entry, EMV, eCheck and tokenized instruments.
- Built the orchestration layer as a thin coordinator over the core authorization engine — composing merchant configuration, BIN lookup, surcharge calculation and tokenized instrument retrieval — keeping business logic outside the PCI-scoped authorization core.
- Implemented merchant-scoped authorization for money-moving endpoints, cross-validating independently-sourced organization identifiers to prevent transactions executing under a merchant the caller isn't entitled to act as.

• MID / TID Management Services

Java, Spring Boot, JPA/Hibernate, MySQL

- Built Merchant ID and Terminal ID provisioning and lifecycle services underpinning payment routing, with REST APIs consumed by onboarding and support tooling.

ACHIEVEMENTS & EDUCATION

Master of Computer Applications — NIT Tiruchirappalli

CGPA: 8.7

Bachelor of Computer Applications — DAVV Indore

CGPA: 6.6